temperature and pressure. Calculate the molar mass of the unknown gas

Solution:

$$\frac{\text{rate}_{\text{unknown}}}{\text{rate } N_2} = \sqrt{\frac{M_{N_2}}{M_{\text{unknown}}}}$$

$$0.5 = \sqrt{\frac{28\ \text{g mol}^{-1}}{M_{\text{unknown}}}}$$

Squaring on both sides

$$(0.5)^2 = \frac{28\ \text{g mol}^{-1}}{M_{\text{unknown}}}$$

$$\Rightarrow M_{\text{unknown}} = \frac{28}{0.25} = 112\ \text{g mol}^{-1}$$

Evaluate Yourself

7. A flammable hydrocarbon gas of particular volume is found to diffuse through a small hole in 1.5 minutes. Under the same conditions of temperature and pressure an equal volume of bromine vapour takes 4.73 min to diffuse through the same hole. Calculate the molar mass of the unknown gas and suggest what this gas might be.

(Given that molar mass of bromine = 159.8 g/mole)

6.5 Deviation from ideal gas behaviour

The kinetic theory of gases (postulates of kinetic theory of gases are described in physics text book) which is the basis for the gas equation (PV=nRT), assumes that the individual gas molecules occupy negligible volume when compared to the total volume of the gas and there is no attractive force between the gas molecules. Gases whose behaviour is consistent with these assumptions under all conditions are called ideal gases. But in practice both these assumptions are not valid under all conditions. For example, the fact that gases can be liquefied shows that the attractive force exists among molecules. Hence, there is no gas which behaves ideally under all conditions. The non-ideal gases are called real gases. The real gases tend to approach the ideal behaviour under certain conditions.

6.5.1 Compressibility factor Z

The deviation of real gases from ideal behaviour is measured in terms of a ratio of PV to nRT. This is termed as compressibility factor. Mathematically,

$$Z = \frac{PV}{nRT}$$

For ideal gases PV = nRT, hence the compressibility factor, Z = 1 at all temperatures and pressures. For these gases the plot of Z vs P should be a straight line parallel to the pressure axis. When a gas deviates from ideal behaviour, its Z value deviates from unity. For all gases, at very low pressures and very high temperature the compressibility factor approaches unity and they tend to behave ideally. The plot of the compressibility factor vs pressure for some common gases are shown in Figure 6.8.

11th Std Chemistry 158-185.indd 169 06-01-2021 15:35:20